

RMC Mix Design Collection – M35 & M40

High-grade concrete records with deliberately different cement and supplementary cementitious-material contents.

M35 Concrete – Mix Design M35-B

Application: Infrastructure and heavy RCC **Target Slump:** 125 mm

Material / Parameter	Quantity	Unit	Remarks
Cement	430	kg/m ³	Primary cementitious binder
Fly Ash	50	kg/m ³	Supplementary cementitious material
Water	158	L/m ³	Before aggregate moisture correction
Fine Aggregate / Sand	675	kg/m ³	Controlled fine aggregate
10 mm Coarse Aggregate	405	kg/m ³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	645	kg/m ³	Nominal coarse aggregate fraction
Chemical Admixture	5.0	kg/m ³	Water-reducing admixture
Water/Cementitious Ratio	0.329	-	Illustrative design value
Target Mean Strength	43.25	MPa	Illustrative 28-day target

Design remarks: Higher-cement M35 alternative. Designed as a separate record so retrieval can distinguish cement content from other M35 designs.

M40 Concrete – Mix Design M40-A

Application: High-rise structural works **Target Slump:** 120 mm

Material / Parameter	Quantity	Unit	Remarks
Cement	420	kg/m ³	Primary cementitious binder
Fly Ash	70	kg/m ³	Supplementary cementitious material
Water	160	L/m ³	Before aggregate moisture correction
Fine Aggregate / Sand	680	kg/m ³	Controlled fine aggregate
10 mm Coarse Aggregate	400	kg/m ³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	650	kg/m ³	Nominal coarse aggregate fraction
Chemical Admixture	4.8	kg/m ³	Water-reducing admixture
Water/Cementitious Ratio	0.327	-	Illustrative design value
Target Mean Strength	48.25	MPa	Illustrative 28-day target

Design remarks: Illustrative M40 formulation with high cementitious content and reduced water.

M40 Concrete – Mix Design M40-B

Application: High-strength concrete with increased binder **Target Slump:** 130 mm

Material / Parameter	Quantity	Unit	Remarks
Cement	460	kg/m ³	Primary cementitious binder
Fly Ash	60	kg/m ³	Supplementary cementitious material
Water	155	L/m ³	Before aggregate moisture correction
Fine Aggregate / Sand	665	kg/m ³	Controlled fine aggregate
10 mm Coarse Aggregate	395	kg/m ³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	645	kg/m ³	Nominal coarse aggregate fraction
Chemical Admixture	5.4	kg/m ³	Water-reducing admixture
Water/Cementitious Ratio	0.298	-	Illustrative design value
Target Mean Strength	48.25	MPa	Illustrative 28-day target

Design remarks: Alternative M40 formulation with significantly higher cement content and lower water.

M40 Concrete – Mix Design M40-C

Application: High-strength pumped concrete **Target Slump:** 135 mm

Material / Parameter	Quantity	Unit	Remarks
Cement	390	kg/m ³	Primary cementitious binder
Fly Ash	110	kg/m ³	Supplementary cementitious material
Water	158	L/m ³	Before aggregate moisture correction
Fine Aggregate / Sand	690	kg/m ³	Controlled fine aggregate
10 mm Coarse Aggregate	400	kg/m ³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	640	kg/m ³	Nominal coarse aggregate fraction
Chemical Admixture	5.1	kg/m ³	Water-reducing admixture
Water/Cementitious Ratio	0.316	-	Illustrative design value
Target Mean Strength	48.25	MPa	Illustrative 28-day target

Design remarks: Alternative M40 formulation using a higher supplementary cementitious-material fraction and lower cement content.

Dataset note: These are synthetic RMC mix-design records created for RAG ingestion, retrieval, comparison, and question-answering tests. They are not approved construction designs. Actual mix proportions must be established and validated using project requirements, material properties, laboratory trials, moisture corrections, and applicable standards.